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A more secure and efficient privacy-preserving collaborative filtering system

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ABSTRACT

Recommendation systems are extensively used to provide personalised services. As a commonly used recommendation algorithm, collaborative filtering (CF) provides predictions of the target user's ratings from the ratings of other users. However, collecting other users' personalised information violates the user's privacy. Although encryption-based privacy-preserving collaborative filtering protocols have been proposed, the existing solutions still suffer from security issues that arise during the recommendation processes and inefficiencies that originated from the techniques used.

Motivated by the deficiencies of the existing protocols, this study aims to investigate an efficient and robust privacy-preserving collaborative filtering protocol. It strives to enhance the performance of recommendation systems while ensuring user privacy. Firstly, the proposed protocol ensures privacy for all participants. Secondly, the proposed protocol achieves better computation efficiency. In contrast to the previous work that used packings and homomorphic rotations to compute the similarities, the proposed protocol uses coefficient embedding and re-encryption. The similarity computation process achieves about 8× more efficiency than the state-of-the-art. However, the space cost of our protocol is somewhat larger. Last but not least, the proposed protocol maintains full rating information, while the previous works only maintain the rounding information of the target user. Theoretically, the proposed protocol provides better accuracy. In addition, the proposed protocol achieves post-quantum security. The security of the proposed protocol is analyzed qualitatively, and quantitative analyses of the computation and communication costs are carried out. The experimental results support the theoretical analysis.

1. Introduction

Recommendation systems are widely used to provide personalized content to the end users, such as movies, books, and others. Collaborative filtering (CF)-based algorithms (Herlocker et al., 1999; Sarwar et al., 2001) are commonly used in recommendation systems. It models the user-item relations from the feedback ratings of the other users. However, gathering other user's personalised information has raised serious privacy concerns. To protect user privacy, privacy-preserving collaborative filtering-based recommendation protocols have been explored.

The previous works Badsha et al. (2016), Erkin et al. (2012b), Kikuchi et al. (2009), Tada et al. (2010) deal with those privacy concerns with the help of homomorphic encryption schemes (Brakerski et al., 2014; Elgamal, 1985; Paillier, 1999). In particular, the ratings are encrypted and then sent to the recommendation server for further processing. From the homomorphic property of the ciphertexts, the server can perform collaborative filtering algorithms over the ciphertexts, and thus data privacy is preserved. However, as we elaborate next, the existing protocols (Badsha et al., 2017; Jumonji et al., 2023) suffer from larger communication and computation overheads and privacy (Alharbe et al., 2023).

The work of Badsha et al. (2017) proposed a privacy-preserving user-based collaborative filtering (PPUBCF) recommendation system from the homomorphic encryption of Boneh et al. (2005). However, there are two main issues: efficiency and security. Considering the efficiency,

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the users encrypt each of their ratings with the BGN scheme (Boneh et al., 2005) and send them to the target user for further computations. Assuming the number of items is M, the target user should perform M multiplications to compute a similarity, which significantly increases its computation overhead and communication overheads. Furthermore, the recommended server also needs to perform many homomorphic computations. The prediction vector is known to the target user from a security aspect, which induces privacy leakage in the other user's interest. In addition, the protocol proposed in Badsha et al. (2017) can be broken in the quantum environment (Shor, 1994).

To achieve post-quantum security, the work of Jumonji et al. (2023) proposed a lattice-based protocol for user-based CF. In the work of Jumonji et al. (2023), the users encrypt their rating vectors with the packing technique of the BGV scheme (Brakerski et al., 2014; Fan & Vercauteren, 2012). In this case, the rating vector can be encrypted to one ciphertext if the number of items is smaller than the packing size. Although the efficiency of Jumonji et al. (2023) is significantly better than that of Badsha et al. (2017), there is still room to improve. In addition, the prediction vector is also exposed to the target user, which is a crucial security issue. Assume an extreme case in which there are only two users (one of which is the target user). The target user can now fully extract the exact rating vector of the other user. Thus, the exposition of the prediction vector is an important security issue for state-of-the-art privacy-preserving CF protocols.

Motivated by the deficiencies of the existing protocols, this study aims to investigate an efficient and robust privacy-preserving collaborative filtering method. It strives to enhance the performance of recommendation systems while ensuring user privacy.

Contributions: This paper proposes a privacy-preserving user-based collaborative filtering protocol. The main distinguishing properties of the proposed protocol are as follows:

- The proposed protocol achieves privacy against the target user. Note that the previous protocols (Badsha et al., 2017; Jumonji et al., 2023) leak the prediction values to the target user. In contrast to the previous works, a random vector additively masked the prediction vector. Then, the additive secret-sharing scheme is implied to address the prediction leakage issues.
- The proposed protocol achieves better computation efficiency. In contrast to the work of Jumonji et al. (2023) that applies packings and homomorphic rotations to compute the similarities, the proposed protocol uses coefficient embedding and re-encryption. As illustrated in Table V, our overall similarity computation process is about 8× more efficient than the SEAL implementation of Jumonji et al. (2023). However, the space cost of our protocol is somewhat larger than that of Jumonji et al. (2023).
- The proposed protocol maintains more user information. Note that the previous work of Jumonji et al. (2023) rounds all the rating vectors (including the target user's ratings) to adapt the BGV scheme. Instead, we simply encrypt the original ratings of the target user. Thus, theoretically, our proposed protocol maintains more user information than that of Jumonji et al. (2023), and thus provides better accuracy.

In addition, As the previous work of Jumonji et al. (2023), the proposed protocol achieves post-quantum security. We implement the proposed protocol to illustrate our theoretical analysis.

2. Related work

The CF-based recommendations are different from content-based recommendations (Balabanovic & Shoham, 1997), it only needs the ratings of the users over the different items. Privacy-preserving CF-based recommendation protocols are mainly based on homomorphic encryption schemes. There are also some related works from other techniques, as we elaborate below.

2.1. Encryption-based approaches

The first privacy-preserving recommender system using ElGamal encryption (Elgamal, 1985) is proposed in the work of Canny (2002). Later, Kikuchi et al. (2009) proposed a privacy-preserving user-based collaborative filtering protocol from the Paillier encryption (Paillier, 1999). However, their protocol suffers from intensive computation and communication costs. Although the work of Tada et al. (2010) reduced both the communication and computation costs of Kikuchi et al. (2009), the similarity values which are generated during the recommendation process are leaked. Erkin et al. (2012a) proposed a privacy-preserving item-based collaborative filtering algorithm (Sarwar et al., 2001) using Paillier encryption (Paillier, 1999). In contrast to the work of Tada et al. (2010), the work of Erkin et al. (2012a) reduced the computational and communication overhead by introducing a Look-up Table. The work of Li et al. (2016) proposed an efficient privacy-preserving item-based CF algorithm which achieves similar or even better recommendation accuracy. The later work of Erkin et al. (2012) further reduced the computational overhead of the user-based collaborative filtering by using packing techniques of homomorphic encryption schemes (Damgård et al., 2007; Paillier, 1999).

To reduce the user-server interaction, the work of Erkin et al. (2012b) introduces a semi-honest third party to assist with completing the secure recommendation. Badsha et al. (2016) enhanced the recommendation performance of Tada et al. (2010) by employing ElGamal encryption (Elgamal, 1985). Later, Badsha et al. (2017) further improved privacy-preserving collaborative filtering efficiency by using BGN encryption (Boneh et al. (2005)). Similar to the work of Erkin et al. (2012b), the work of Badsha et al. (2017) needed a semi-honest third party to help the recommendation server to decrypt the perturbed predictions. Thus, their construction didn't leak the similarity to the target user. The latest work of Jumonji et al. (2023) proposed a privacy-preserving user-based CF protocol, called BGV-CF, using the fully homomorphic encryption scheme (Brakerski et al., 2014). The packing technique reduces the number of interactions and communication between the user and the server compared with the work of Badsha et al. (2017). In addition, the computational overhead of the recommendation phase is reduced by offloading some computation of the recommendation phase to the user. However, the work of Jumonji et al. (2023) has some deficiencies, as stated in the introduction.

2.2. Others

The work of Polat and Du (2003) proposed a privacy-preserving CF from the randomized perturbation techniques, where the user ratings are perturbed, which reduces the accuracy of the recommendation. The later work by Zhang et al. (2006b) improves the accuracy deficiency of Polat and Du (2003) by more interaction with the recommendaer server. McSherry and Mironov (2009) studied the CF-based algorithms from differerencial privacy that the ratings are perturbed by bounded distributions. Although the performance of the perturbed-based CF algorithsms are very efficent, their accuracy is concerning. In addition, the user privacy could be leaked (Huang et al., 2005; Zhang et al., 2006a). The work of Yang et al. (2024) proposed a privacy-preserving multi-behaviour recommendation for heterogeneous one-class collaborative filtering exploiting federated learning (McMahan et al., 2017). The work of Polatidis et al. (2017) proposed a multi-level privacy-preserving solution for collaborative filtering systems via perturbing each rating before they are submitted to the server. Alharbe et al. (2023) studied the collaborative filtering algorithm via combining the embedding

Table 1
Frequently used notations.

Notations	Description
N	Number of users
M	Number of items
n	dimension of the polynomial ring
$[k]$	Set of integers $1, 2, \dots, k$
u_i, C-user	The i-th common user
u_t, T-User	The target user
U	Set of users
$\vec{r}_i$	The rating vector of u_i
$\vec{R}$	The rating matrix
P	The prediction vector
P_i	The prediction value of ith item
RS	Recommendation server
pk	Public keys of FHE scheme
sk	Private keys of FHE scheme
Enc, Dec	Encryption, decryption algorithm
γ	Amplification factor
$\times_p$	packed multiplication
$\vec{1}$	All 1 vector of length n
$\oplus$	homomorphic addition
$\otimes$	homomorphic multiplication
$\otimes_p$	homomorphic packed multiplication

representation and word embedding techniques. Kaya and Kaleli (2022) proposed an intuitionistic fuzzy set-based top-n recommendation system with a neighborhood formation process for multi-criteria datasets.

3. Preliminaries

Notations

The frequently used notations in this paper are listed in Table 1.

3.1. User-based collaborative filtering

A popular method for creating recommender systems is User-Based Collaborative Filtering (UBCF), which makes predictions about what a target user will want based on the ratings and similarity of other users. A rating is a non-negative integer that expresses a user's preference for an item. Let $U := \{u_1, u_2, \dots, u_N\}$ be the set of N users and $\boldsymbol{R} := \{\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N\} \in \mathbb{Z}^{N\times M}$ be the rating matrix,[1] then the aim of UBCF is to predict the ratings that a target user, denoted by u_t, will assign to the unrated items. Let $P := [P_1, P_2, \dots, P_M] \in \mathbb{R}^M$ be the rating predictions (may be not an integer valued vector), then the jth prediction is given as follows:

$$P_j = \frac{\sum_{i\in[N]} d(u_t, u_i) \cdot \vec{r}_{i,j}}{\sum_{i\in[N]} d(u_t, u_i)}, \tag{1}$$

where $d(u_t, u_i)$ is the cosine distance between the target user u_t and user u_i. After computing the prediction vector P, the top predicted items, that are didn't yet be rated by the target user, are recommended. However, we will use a subtile reformulated prediction as stated in the main body of the paper.

3.2. Fully homomorphic encryption

There's no doubt about it: fully homomorphic encryption (FHE) is a form of encryption that allows one to perform computations over the encrypted data without needing to decrypt it first. This guarantees that data will remain encrypted and secure throughout its life cycle, including during processing. More precisely, for the FHE encryption algorithm, denoted Enc, the following holds :

$$\mathsf{Enc}(m_1) \oplus \mathsf{Enc}(m_2) = \mathsf{Enc}(m_1 + m_2)$$
$$\mathsf{Enc}(m_1) \otimes \mathsf{Enc}(m_2) = \mathsf{Enc}(m_1 * m_2),$$

where $\oplus$ and $\otimes$ are the homomorphic operations. In addition, we use an important tool – packing. The packing technique of the FHE scheme enables the user to perform operations on message vectors simultaneously. More precisely, for two vectors $\vec{m} = (m_1, \dots, m_n)$ and $\vec{m}' = (m'_1, \dots, m'_n)$,and $\mathsf{ct} := \mathsf{pack} \cdot \mathsf{Enc}(\vec{m})$, $\mathsf{ct}' := \mathsf{pack} \cdot \mathsf{Enc}(\vec{m}')$

$$\mathsf{ct} \oplus \mathsf{ct}' = \mathsf{pack} \cdot \mathsf{Enc}(m_1 + m'_1, \dots, m_n + m'_n)$$
$$\mathsf{ct} \otimes \mathsf{ct}' = \mathsf{pack} \cdot \mathsf{Enc}(m_1 \cdot m'_1, \dots, m_n \cdot m'_n)$$

where $\mathsf{pack} \cdot \mathsf{Enc}$ is the packed homomorphic encryption. Although the packing technique provides simultaneous operations on the messages, the operations only adapted over the elements at the same position. Thus, the cross-position operations are very inefficient.

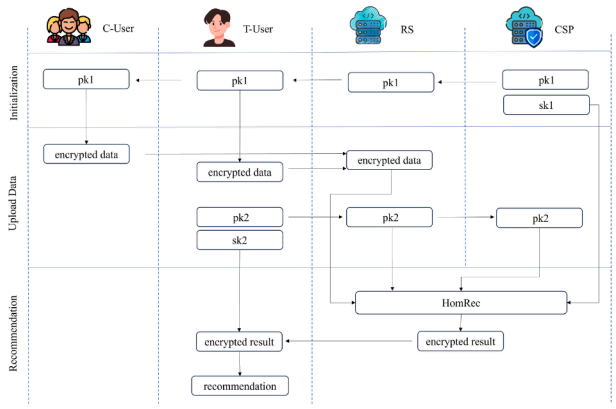


Fig. 1. High-level overview of a recommendation System.

3.3. Additive secret sharing

In the case of a 2-party system, we define the additive shares of x as $[\![x]\!]_1$ and $[\![x]\!]_2$, provided that $[\![x]\!]_1 + [\![x]\!]_2 = x$. It is straightforward to see that the secret shares support secure additions. Furthermore, the process is non-interactive and can be achieved locally, which is a significant advantage. Furthermore, the secure multiplication of a constant with the given additive shares is also performed locally. However, to perform the multiplication of two shares, the use of Beaver's triples is necessary. Since the additions and multiplications are performed given the additive shares, complex operations and arithmetic circuits can be evaluated. Therefore, additive shares provide secure computation for mutually distrusted participants, maintaining the integrity and privacy of inputs even in the presence of dishonest behaviors. Several methods exist for implementing additive share-based secure evaluations. In our work, we utilize LSSS within the SPDZ (Keller, 2020) framework to facilitate distance comparisons while ensuring the security of location privacy.

4. The problem formulation

4.1. System model

The system consists of a recommendation server (RS) and N Users and a cryptographic service provider (CSP). Additionally, there is a user who wants to get a recommended item. We refer this user as target user and denote as u_t. Based on the other users' preferences, some candidate items are recommended to u_t. The CSP is the provider of cryptographic services such as generating keys for the FHE scheme and providing decryption of FHE ciphertexts. In addition, the CSP may involve

[1] $\vec{r}_i \in \mathbb{Z}^M$ denotes the rating vector of the ith user for M items, and $\vec{r}_{i,j}$ is the rating of the u_i for the jth item. The rating $\vec{r}_{i,j} \geq 1$ if the item is purchased before, and $\vec{r}_{i,j} = 0$, otherwise.

some secure computations. The RS is responsible for providing computational and storage resources for the recommendation process. The communications and functionality of each participants are shown in Fig. 1.

As shown in Fig. 1 The recommendation protocol consist of following phases: Initialization, UpdateData and Recommendation phase.

Initialization: In this phase, the CSP generates public and private keys of the fully homomorphic encryption scheme and sends the public key to the RS, and the RS forwards these keys to all users.

Upload Data: In this phase, the target user generates its own public secret key pair and then encrypts its rating vector with the FHE key (received from the RS) and sends the ciphertext and the its public key to the RS. The other users also encrypt their ratings with the public key and send it to the RS.

Recommendation: After receiving the encrypted rating vectors, the RS first to homomorphically compute the prediction vector. Then, the RS jointly selects the best possible predictions, and then securely sends the result to the target user.

At the end, the target user gets the index/name of the top k unrated items as the recommendation.

4.2. Security assumptions

The above recommendation protocol is designed to protect user privacy and ensure the overall security of the protocol by making several key security assumptions.

- **Secure Communication:** The communication channels between the users, RS, and CSP are assumed to be secure. Since the SSL protocols provide a secure communication channel, preventing eavesdropping and tampering, this type of assumption is plausible.
- **Collusion:** The threat model assumes semi-honest participants won't collude with others. It assumes the CSP won't collude with users, and RS won't collude with CSP. Similarly, it assumes common users and target users won't collude with each other.

4.3. Threat models and design goals

This paper presents a system that employs a semi-honest security model, wherein the user and server collaborate to complete recommendations. Users and RS follow the protocol and are eager to obtain private information. All communication channels between users and RS are secure. This paper examines the data privacy of user inputs, intermediate results, and outputs.

Threat models. The following threat models outline the roles and behaviors of the various participants, emphasizing potential risks to user privacy and data security.

Semi-honest CSP: In this context, the CSP functions without any malevolent intent and in strict accordance with the established protocol. Nevertheless, it may endeavor to procure further data regarding the prediction and rating values.

Semi-honest Users: In this scenario, users adhere to the established protocols but may attempt to discern information about other users or the system's internal operations. Additionally, users may endeavor to obtain further insights into the target user's ratings.

Semi-honest Target User: The target user may update their encrypted rating vectors, but may also attempt to glean information about the other user, including ratings and prediction vectors.

Semi-honest RS: In this configuration, the recommendation processes requests and performs matching according to the protocol, but may also seek to learn additional information about users based on the interactions with the CSP and the encrypted rating vector's ciphertexts.

Design goals . This research aims to enhance the performance of privacy-preserving recommender systems while reducing communication costs. To do so, we must address the following research challenges.

- **Efficiency:** The protocol's design enhances computational and communication efficiency. It employs sophisticated encryption methodologies, including FHE and ciphertext packing, to minimize data exchange and processing times.
- **Privacy Protection:** The protocol's security goal is to maintain the privacy of all user's ratings. To achieve this, the encryption schemes and additive secret sharing schemes are applied.
- **Practicality:** The recommendation protocol is designed with practicality in mind, with a special emphasis on low requirements for mobile devices. Because users only need to store ratings, the system is simple to set up and operate. This approach is practical for both users and service providers because it is easy to use and guarantees robustness in a variety of situations.

The suggested protocol that satisfies the aforementioned design objectives and offers security in the aforementioned threat models is described in detail in the sections that follow.

5. Our recommendation protocol

In this section, we first show the detailed design of our UBCF-based recommendation protocol. Then, we theoretically analysis the proposed protocol.

5.1. The detailed design

The recommender system protocol consist of three phase: Initialization, Upload Data and Recommendation phases. The detailed description of each phases are as follows.

Initialization. In this phase, the CSP runs the FHE key generation algorithm to generate FHE key pair as follows:

$$(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{FHE} \cdot \mathsf{keyGen}(\lambda),$$

where λ is the security parameter. Then the CSP sends the public key to the RS to publish it. Thereafter, the RS will proxy it to the users. This process is illustrated in Fig. 2.

Upload Date. In this phase, each user u_i holds a rating vector $\vec{r}_i := \{r_{i,1}, r_{i,2}, \ldots, r_{i,M}\}$. For each $r_{i,j}$, it's value is an integer between 1 and $r_{\max}$ if $r_{i,j}$ has been rated by u_t before. Otherwise, it is set to 0. To upload rating vectors, each common user u_i runs Algorithm 1 as follows:

$$\mathsf{ct}_i, \mathsf{ct}_i^n \leftarrow \mathsf{ComDataEnc}(\vec{r}_i, \gamma), \tag{2}$$

where γ is some large enough amplification factor. Then, the user u_i sends the $\mathsf{ct}_i, \mathsf{ct}_i^n$ to the recommendation server. The target user u_t runs Algorithm 2 as

$$\mathsf{ct}, \mathsf{ct}^{\vec{P}} \leftarrow \mathsf{TarDataEnc}(\vec{r}_t), \tag{3}$$

where $\vec{r}_t$ is the rating vector of the target user. Next, the target user u_t generates public and private key pair $(\mathsf{pk}_t, \mathsf{sk}_t)$ of an PKE scheme, and sends pk_t, $\mathsf{ct}_t, \mathsf{ct}_t^{\vec{P}}$ and an integer k to the recommendation server to get k recommendation from the RS.

Remark 5.1. Note that the previous work of Jumonji et al. (2023) rounds the rating vector of the target user. However, as shown in Algorithm 2, we simply encrypt the original ratings of the target user. Thus, theoretically, our proposed protocol maintains more user information than that of Jumonji et al. (2023).

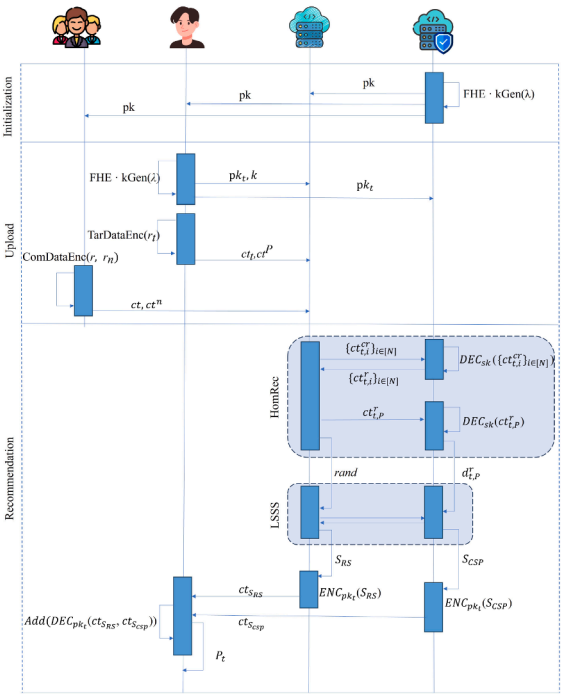


Fig. 2. Our recommendation system.

Algorithm 1 $\mathsf{ComDataEnc}(\vec{r}, \vec{r}^n) \to \mathsf{ct}, \mathsf{ct}^n$.

Input: a rating vector $\vec{r}$ and its normalized vector $\vec{r}^n$.
Output: ciphertexts $\mathsf{ct}, \mathsf{ct}^T$

$\mathsf{ct} := \mathsf{pack} \cdot \mathsf{Enc}_{\mathsf{pk}}(\vec{r})$
For $i = 1$ to M Do
  $\vec{r}_i^n := \left[\vec{r}_i \cdot \gamma / \sqrt{\sum_{k=1}^{M} r_k^2}\right]$
End For
$r^n(x) := \sum_{i\in[M]} \vec{r}_i^n \cdot x^i$
$\mathsf{ct}^n := \mathsf{Enc}_{\mathsf{pk}}(r^n(x))$
Output: $\mathsf{ct}, \mathsf{ct}^n$

Algorithm 2 $\mathsf{TarDataEnc}(\vec{r}_t) \to \mathsf{ct}_t, \mathsf{ct}^{\vec{P}}$.

Input: a rating vector $\vec{r}_t$.
Output: ciphertexts $\mathsf{ct}_t, \mathsf{ct}^{\vec{P}}$

Let $r_t(x) := \sum_{i\in[N]} \vec{r}_{t,i} \cdot x^{n-i}$
$\mathsf{ct}_t := \mathsf{Enc}_{\mathsf{pk}}(r_t(x))$
For $i = 1$ to M Do
  Let $\vec{v}_{t,i} = 0$
  If $\vec{r}_{t,i} = 0$ **then**
    $\vec{v}_{t,i} = 1$
  End If
End For
$\mathsf{ct}^{\vec{P}} := \mathsf{pack} \cdot \mathsf{Enc}_{\mathsf{pk}}(\vec{v}_t)$
Output: $\mathsf{ct}_t, \mathsf{ct}^{\vec{P}}$

Recommendation. In this phase, after receiving the encrypted rating vectors, the RS transmits pk_t to the CSP, and cooperate with the CSP to run Algorithm 3 to get the final indexes of the recommended items. To this, they runs

$$\vec{rand}, d^r_{t,\vec{P}} \leftarrow \mathsf{HomRec}\left(\{\mathsf{ct}_i, \mathsf{ct}_i^n\}_{i\in[N]}, (\mathsf{ct}_t, \mathsf{ct}_t^{\vec{P}}), k\right), \tag{4}$$

where $\vec{rand}$ is a random integer vector that owned at RS side; $d^r_{t,\vec{P}}$ is a ring element (a integer vector) that achieved at CSP side. Then, the RS and the CSP cooperatively runs the LSSS protocol to obtain a additive shares of the indexes of the recommended items. That is, they run

$$\vec{s}_{RS}, \vec{s}_{CSP} \leftarrow LSSS(\vec{rand}, d^r_{t,\vec{P}}, k) \tag{5}$$

where the vectors $\vec{s}_{RS}, \vec{s}_{CSP}$ are the additive shares such that $\vec{s}_{RS} + \vec{s}_{CSP}$ is the indexes of the recommended items. That is the ith entry of $\vec{s}_{RS} + \vec{s}_{CSP}$ is 1 if ith item is recommended, and is 0 if not recommended. Note that there are at most k places are 1 in $\vec{s}_{RS} + \vec{s}_{CSP}$.

Algorithm 3 $\mathsf{HomRec}\left(\{\mathsf{ct}_i, \mathsf{ct}_i^n\}_{i\in[N]}, (\mathsf{ct}_t, \mathsf{ct}^{\vec{P}}), k\right) \to \vec{rand}, d^r_{t,\vec{P}}$.

Input Rating vectors $\{\mathsf{ct}_i, \mathsf{ct}_i^c\}_{i\in[N]}, (\mathsf{ct}_t, \mathsf{ct}_t^{\vec{P}})$ and an integer k.
Output: $\vec{rand}, d^r_{t,\vec{P}}$

/* The RS Runs as follow: */
For $i = 1$ to N Do
  $\mathsf{ct}^n_{t,i} := \mathsf{ct}_i^n \otimes \mathsf{ct}_t$
  Sample a random vector $\vec{rand}_i \leftarrow \mathbb{Z}_p^n$
  Let $rand_i(x) := \sum_{j\in[n]} \vec{rand}_{i,j} \cdot x^j$
  $\mathsf{ct}^{cr}_{t,i} := \mathsf{ct}^n_{t,i} \oplus \mathsf{Enc}(-rand_i(x))$
End For
RS sends $\{\mathsf{ct}^{cr}_{t,i}\}_{i\in[N]}$ to the CSP
/* The CSP runs the following: */
For $i = 1$ to N Do
  $d^r_{t,i} := \mathsf{Dec}_{\mathsf{sk}}(\mathsf{ct}^{cr}_{t,i})$
  let $\vec{d}^r_{t,i} := d^r_{t,i}[n] \cdot \vec{1}$ ▷ $\vec{1} \in \mathbb{Z}^n$ is the all 1 vector
  $\mathsf{ct}^r_{t,i} := \mathsf{pack} \cdot \mathsf{Enc}_{\mathsf{pk}}(\vec{d}^r_{t,i})$
End For
CSP sends $\{\mathsf{ct}^r_{t,i}\}_{i\in[N]}$ back to the RS
/* The RS runs the following: */
For $i = 1$ to N Do
  $\mathsf{ct}'_{t,i} := \mathsf{ct}^r_{t,i} \oplus_p \mathsf{Enc}(\vec{rand}_i[n] \cdot \vec{1})$
  $\mathsf{ct}_{t,i} := \mathsf{ct}_i \otimes_p \mathsf{ct}'_{t,i}$
End For
$\mathsf{ct}_{t,all} := \bigoplus_{i\in[N]} \mathsf{ct}_{t,i}$
$\mathsf{ct}_{t,\vec{P}} := \mathsf{ct}_{t,all} \otimes_p \mathsf{ct}^{\vec{P}}$
$\vec{rand} \leftarrow \mathbb{Z}_p^n$
$\mathsf{ct}^r_{t,\vec{P}} := \mathsf{ct}_{t,\vec{P}} \oplus_p \mathsf{pack} \cdot \mathsf{Enc}(-\vec{rand})$
RS sends $\mathsf{ct}^r_{t,\vec{P}}$ to the CSP
/* The CSP runs the following: */
$d^r_{t,\vec{P}} := \mathsf{Dec}_{\mathsf{sk}}(\mathsf{ct}^r_{t,\vec{P}})$
Output: return $rand$ to RS; return $d^r_{t,\vec{P}}$ to CSP.

Thereafter, the CSP and RS encrypts the corresponding shares with the public key pk_t of the target user. The CSP sends the ciphertext $\mathsf{ct}_{\vec{s}_{CSP}} := \mathsf{Enc}_{\mathsf{pk}_t}(\vec{s}_{CSP})$ of the shares to the target user through the RS, and the RS sends the ciphertext $\mathsf{ct}_{\vec{s}_{RS}} := \mathsf{Enc}_{\mathsf{pk}_t}(\vec{s}_{RS})$ to the target user as well. Finally, after receiving the ciphertexts $\mathsf{ct}_{\vec{s}_{CSP}}, \mathsf{ct}_{\vec{s}_{RS}}$, the target user first to decrypt them and then add them to get the index vector $\vec{P}_t$ of the recommended items. Below, we analysis the proposed protocol.

5.2. Analysis of the proposed protocol

In this section, we present the correctness of the proposed protocol. Then we show the security and the theoretical performance of the protocol.

Correctness. To show the correctness, we need to show that the non-zero entries in the predicted vector $\vec{P}_t$ is corresponds to the largest top k UBCF-based recommendation predictions. From the description of the

prediction vector $\vec{P}$ in Eq. (1), we know that

$$\vec{P}_j = \frac{\sum_{i\in[N]} \frac{\vec{r}_i^\top \cdot \vec{r}_t}{\|\vec{r}_i\|\cdot\|\vec{r}_t\|} \cdot \vec{r}_{i,j}}{\sum_{k\in[N]} \frac{\vec{r}_k^\top \cdot \vec{r}_t}{\|\vec{r}_k\|\cdot\|\vec{r}_t\|}} = \frac{\sum_{i\in[N]} (\frac{\vec{r}_i^\top}{\|\vec{r}_i\|} \cdot \vec{r}_t) \cdot \vec{r}_{i,j}}{\sum_{k\in[N]} \frac{\vec{r}_k^\top \cdot \vec{r}_t}{\|\vec{r}_k\|}}. \tag{6}$$

In addition, the term $\sum_{k\in[N]} \frac{\vec{r}_k^\top \cdot \vec{r}_t}{\|\vec{r}_k\|}$ is deterministic and independent of the indexes i and j. Therefore, the prediction vector $\vec{P}$ is proportional to the term $\sum_{i\in[N]} (\frac{\vec{r}_i^\top}{\|\vec{r}_i\|} \cdot \vec{r}_t) \cdot \vec{r}_{i,j}$. Since the 2-norm of the rating vector $\vec{r}_i$ is a real number, and thus incompatible with homomorphic encryption schemes. Thus, following the previous works Badsha et al. (2017), Jumonji et al. (2023), the 2-norm of $\vec{r}_i$ is first multiplied by an amplification factor γ, and then rounded down to the nearest integer. That is we use the following pseudo-normalization $\vec{r}_i''$ of $\vec{r}_i$ as $\|\vec{r}_i\| := \left\lfloor \vec{r}_i \cdot \gamma / \sqrt{\sum_{k=1}^{M} r_k^2} \right\rfloor$. The effect of this rounding method is negligible for the large enough γ. The work of Jumonji et al. (2023) note that setting $\gamma = 100,1000$ is enough for UBCF-based recommendations. And thus, finding the indexes of the top-k largest prediction values in $\vec{P}$ is equivalent to find the indexes of top-k prediction values of $\vec{P}' = [\vec{P}'_1, \ldots, \vec{P}'_M]$, where we set

$$\vec{P}'_j := \sum_{i\in[N]} (\vec{r}_i''^\top \cdot \vec{r}_t) \cdot \vec{r}_{i,j}. \tag{7}$$

As the setting of $\vec{v}_t$ in Algorithm 2 is the one-hot encoding of the unrated items of the target user, $\vec{P}' \times_p \vec{v}_t$ is the rating-predictions of the unrated items of target user. Hence, from the Eqs. (4) and (5), if $\vec{s}_{RS}$ and $\vec{s}_{CSP}$ are the additive shares of $\vec{P}' \times_p \vec{v}_t$, then the top-k largest elements of $\vec{P}' \times_p \vec{v}_t$ are match with the non-zero elements of $\vec{P}_t$, and thus the correctness of the proposed protocol is holds. The following theorem shows this fact. Due to the space limit, the proof of the following theorem is defer to the supplementary materials 8.

Theorem 5.1. *Let $\{\mathsf{ct}_i, \mathsf{ct}_i^n\}_{i\in[N]}$ are generated by running Algorithm 1 and $(\mathsf{ct}_t, \mathsf{ct}^{\vec{P}})$ is generated by running Algorithm 2, then the outputs $\vec{s}_{RS}$ and $\vec{s}_{CSP}$ of Algorithm 3 are the additive shares of $\vec{P}' \times_p \vec{v}_t$.*

From the above discussion, we know that the proposed protocol can correctly implements the UBCF-based recommendation. Below, we show the security analysis.

Security. As we states in the previous section, the users, the RS and the CSP are semi-honest. Although, they may follow the protocol script, they may extract more information from the received transcripts. Below, we show that, these entities learn nothing except they deserved to learn. To show this, we use the indistinguishability-based technique. Let $\mathcal{A}^{\Pi}$ be the view of an adversary $\mathcal{A}$ in the real recommendation protocol Π, $\mathcal{A}^{\mathsf{Ideal}}$ be the view of an adversary $\mathcal{A}$ in the ideal recommendation protocol Ideal, then the security definition is as follows.

Definition 1. We call a recommendation protocol is secure against a semi-honest probabilistic polynomial time (PPT) adversary $\mathcal{A}$ if following holds.

$$\mathcal{A}^{\Pi} \approx \mathcal{A}^{\mathsf{Ideal}}.$$

Furthermore, if the adversary learns an additional information $f(\vec{x})$ of the input $\vec{x}$, then we call the protocol is secure with the leakage of $f(x)$.

From the description of the recommendation protocol, the common users only updates their encrypted rating vectors. Therefore, they learn nothing about the other user's (including the target user) ratings or preferences. Additionally, the target user only receives the encrypted shares of the final recommendation indexes. Thus, from the security of the PKE scheme that used to encrypt the additive shares and underlying LSSS, we have following result.

Theorem 5.2. *The proposed protocol is secure agains semi-honest users. Additionally, it is secure agains semi-honest target user with the leakage of the indexes of the recommended items if the PKE and the LSSS is secure.*

For the semi-honest RS, and CSP, we have following security results.Due to the space limit, proofs of the following theorems are defer to the supplementary materials 8.

Theorem 5.3. *The proposed protocol is secure against semi-honest RS if the FHE scheme is IND-CPA secure and the LSSS is secure.*

Theorem 5.4. *The proposed protocol is secure against semi-honest CSP if the LSSS is secure.*

Above, we have showed that the proposed protocol can achieve security against the all type of threats we have state in the previous section. Below, we analysis the theoretical performance of the protocol.

Performance. From the description of the protocol, in the initialization phase, the CSP runs an FHE key generation process. In the data updation phase, the common users are need to encrypt their rating vectors and the psudo-normalized vectors as in Algorithm 1. The target user needs to encrypt its rating vector and the one-hot encoding of the its unrated items. Thus the computation cost of the users is $2 \cdot \mathsf{Enc}$ while the communication cost is $2 \cdot \mathsf{ct}$. For the recommendation phase, from the description of the Algorithm 3, we know that the RS needs to run $N\times$ homomorphic multiplications and $N\times$ plain additions (here we omit the uniform sampleing complexity due to its cost is negligible comparing with the homomorphic oprations) for the step 1 to step 6. In addition, for the step 12–15, the RS runs N plain additions plus N homoomrphic multiplications. For the step 16, it runs $N-1$ homomorphic additions. Furthermore, the RS one additional homomorphic multiplication and an plain addition for step 17 and step 19 of Algorithm 3. At the CSP side, the CSP runs N decryption and encryption for step 7–11, and one decryption for the last step of Algorithm 3. In addition, the RS and CSP runs LSSS to get the additive shares of the recommended item's indexes. Finally, they run an PKE encryption to encrypt the shares. The computation and communication costs are listed in Table 2, where we only list the FHE operations and PKE operations due to they are the most consuming part of the protocol.

Table 2 shows the high-level performance overview of the proposed protocol. In the next section, we show the concrete performance of the proposed protocol.

6. Performance evaluation

Here, we evaluate the performance of our protocol. We first describe our experimental settings. Then, we present the experimental results and conduct a comparative analysis with existing methods.

6.1. Experimental settings

The experiments are conducted on a laptop running Ubuntu 22, equipped with an Intel i7 processor at 2.6 GHz and 16 GB of RAM. Unless otherwise specified, all experiments are executed using a single thread.

Dataset. We implement our protocol on the MovieLens dataset (Harper & Konstan, 2015), which is widely used in recommendation system designs. The dataset includes user-generated movie ratings, and each rating takes values in $\{0,1,2,3,4,5\}$.

Cryptographic settings. We use the BGV scheme of Brakerski et al. (2014) to instantiate the FHE, implemented in SEAL library (Chen et al., 2017). The polynomial degree and the scheme modulus are set to 8192 and 133857281. The LSSS scheme is implemented in the MP-SPDZ library of Keller (2020) with four threads. All experiments were performed at 128-bit security level.

Table 2
The computation and communication costs of the proposed protocol.

Process / Complexity	Initialization	Update Data		Recommendation		
	CSP	T-User	C-User	RS	CSP	T-User
Computation	$1 \cdot \mathsf{keyGen}$	$2 \cdot \mathsf{Enc} + 1 \cdot \mathsf{PKE} \cdot \mathsf{keyGen}$	$2 \cdot \mathsf{Enc}$	$(2N+1) \cdot Mul + (2N+1) \cdot PlainAdd + (N-1) \cdot Add + TransLSSS_{RS}^{comp}$	$(N+1) \cdot \mathsf{Dec} + N \cdot \mathsf{Enc} + TransLSSS_{CSP}^{comp}$	$1 \cdot \mathsf{PKE} \cdot \mathsf{Dec}$
Communication	pk	$2 \cdot \mathsf{ct} + 1 \cdot \mathsf{pk}_t$	$2 \cdot \mathsf{ct}$	$(N+1) \cdot \mathsf{ct} + 2 \cdot \mathsf{PKE} \cdot \mathsf{ct} + TransLSSS_{CSP}^{comm}$	$N \cdot \mathsf{ct} + 1 \cdot \mathsf{PKE} \cdot \mathsf{ct} + TransLSSS_{CSP}^{comm}$	–

The communication cost refers to the data size of sent by the participants. $\mathsf{keyGen}, \mathsf{Enc}, \mathsf{Dec}$ refer to the FHE key generation, encryption and decryption algorithms; Mul, $PlainAdd, Add$ denote the multiplication of two FHE ciphertext, addition of an FHE ciphertext with a plaintext and homoomrphic addition algorithm; $\mathsf{PKE} \cdot \mathsf{keyGen}, \mathsf{PKE} \cdot \mathsf{ct}, \mathsf{pk}_t$ denote the key generation algorithm, a ciphertext and the public key of the PKE scheme; The $TransLSSS_{\mathcal{A}}^{comp}, TransLSSS_{\mathcal{A}}^{comm}$ denotes the communication and computation cost of party $\mathcal{A}$ during finding the top-k largest elements from the given additive shares of a vector.

Table 3
The costs of the initialization and data update phases.

Costs / Process	Initialization	Update Data	
	CSP	T-User	C-User
Computation(ms)	1.77	8.18	8.18
Communication(KB)	529	848	848

Protocol Parameters. The number of users takes values in $\{50, 100, 150, 200\}$, while the movie items take values in $\{500k | k \in [10]\}$.

6.2. Experimental results and analysis

Note that the computation and communication overheads of each user depend on the underlying homomorphic encryption scheme as well as the number of items. If the number of items is smaller than the packing size of the FHE scheme, then the running time and the communication overheads only depend on the efficiency of the FHE scheme. In our experiment, we set the number of items as the previous work of Jumonji et al. (2023), which is smaller than the packing size. Thus, the computation and communication overheads remain constant for different numbers of items. Table 3 shows the performance evaluations in the Initialization and Data Update phases. Note that the users' costs (including the target user) are small and can be offline. In addition, they can be done once and for all.

The protocol description shows that the RS performs $2N+1$ homomorphic multiplications and other $O(N)$ plain and homomorphic additions. In addition, RS and CSP perform many comparisons to find the k-top predictions. Thus, the performance of the RS relies on the number of users, the number of items, and the recommendation number. Fig. 3 shows the performance of the RS and CSP during the recommendation phase. Fig. 3(a) and (b) show the performance of the RS and the CSP for running Algorithm 3. Observe that the computation and communication costs for the RS and CSP increase with the number of users. This is the case since the RS evaluates a homomorphic multiplication for each user, and the CSP performs an FHE re-encryption for each similarity computation. Thus, the number of homomorphic multiplications and re-encryptions is linear in terms of the number of users. Since those operations are the most time-consuming part in Algorithm 3, the time costs are linearly increased with the number of users. However, the overall cost is still better than the previous works, as shown in the next section. In addition, at least two ciphertexts will be transmitted between the RS and the CSP to compute the predictions homomorphically. Thus, the communication overheads are also linear in the number of users.

Note that the top-k predictions can be achieved by recursively running the algorithm of finding top-1. Thus, the performance of the top-1 case can roughly illustrate the performance of top-k. Fig. 3(c) and (d) show the performance of the RS and CSP when finding the Top-1 predictions. It is easy to see that the time and communication costs are linear in the number of items. This is because the secure comparison is the most time and space-consuming part of finding the top-1 element. Another observation is that the finding of the top-k process is independent of the number of users.

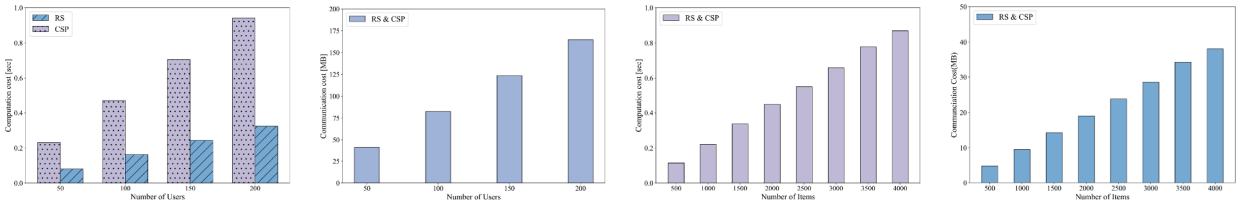
Note that the computation and communication costs of the recommendation phase are obtained from the computation and communication costs in Algorithm 3 plus the evaluation costs generated in finding the top-k predictions. Additionally, the total performance of the protocol for making a secure user-based recommendation is the sum of the costs of all participants. Fig. 4 shows the computation and communication costs of the protocol for Top-1 recommendation. Fig. 4(a) shows the time cost of the protocol with the number of users and items. It is not hard to see that the time cost of the protocol increases with the number of users and items. Another observation is that the number of users impacts the efficiency more than the number of items. This is because the homomorphic multiplications and re-encryptions are more time-consuming than comparisons. Fig. 4(b) shows the space cost of the protocol with the number of users and items. Observe that the time cost of the protocol increases with the number of users and items. However, the increment of users is much stronger than the number of items. This is because the homomorphic ciphertexts are much larger than the transactions generated during the comparisons. Below, we compare our protocol with the state-of-the-art protocols.

6.3. Comparisons

The previous works Badsha et al. (2017), Jumonji et al. (2023) also studied the privacy-preserving collaborative filtering-based recommendation with homomorphic encryptions. However, they didn't protect the prediction vector from the target user. In addition, the protocol of Badsha et al. (2017) is not secure in the quantum environment (Shor, 1994), while the proposed protocol and the protocol of Jumonji et al. (2023) achieve post-quantum property. The high-level security comparisons are listed in Table 4.

Note that the work of Badsha et al. (2017) used BGN scheme (Boneh et al., 2005) to encrypt each rating. As Badsha et al. (2017) stated, their protocol takes several hours and several GB communication overheads to perform a user-based collaborative filtering-based recommendation for 200 users and 500 items. In addition, their protocol is not a post-quantum protocol. Therefore, we didn't list it in the following efficiency comparisons.

Note that the work of Jumonji et al. (2023) also used packing techniques of the BGV scheme in HElib (Halevi & Shoup, 2020). Firstly, we evaluate the BGV-CF experiment of Jumonji et al. (2023) on HElib library with the same parameter settings in there. Furthermore, to be fair,



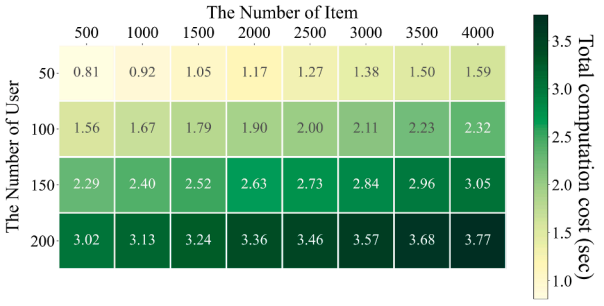
(a) Time cost for Alg.3. (b) Communication cost for Alg.3. (c) Time cost for Top-1 predictions. (d) Communication cost for Top-1 predictions.

Fig. 3. Performance of RS and CSP in homomorphic recommendation phase.

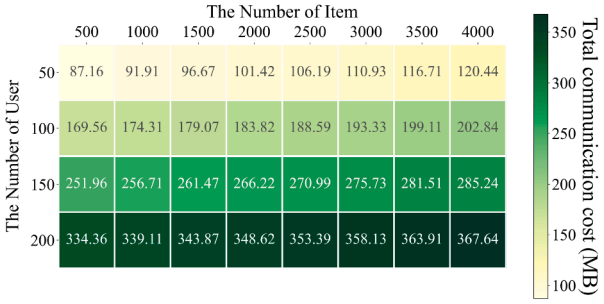
Table 4
Comparison among the privacy-preserving CF-based recommendation protocols.

Solutions	NoLeak-Server	NoLeak-TUser	NoLeak-CUser	Post-quantum	Techniques
Badsha et al. (2017)	✓	✗	✓	✗	Plain FHE
Jumonji et al. (2023)	✓	✗	✓	✓	Packing
Ours	✓	✓	✓	✓	pakcing + coeffs

We use ✓(✗) to represent if the protocol achieved the corresponding property. The columns "NoLeak-Server", "NoLeak-TUser", and "NoLeak-CUser" denote that the prediction vector is not known to the server, target user and common users. "Post-quantum" denotes that the underlying protocol is post-quantum. "packing" and "coeffs" denotes that the packing technique of the HE scheme and coefficient embedding is used.



(a) Time cost of the homomorphic evaluation phase.



(b) Communication cost for Top-1 predictions.

Fig. 4. Performance of the proposed protocol.

we also evaluate their protocol using the SEAL library to eliminate the impact of FHE instantiations on the comparison itself. As the work of Jumonji et al. (2023) suggested, the parameters are also set (as ours) such that the number of packing slots is larger than the number of items. In addition, we use a single thread for our protocol and the baseline protocol of Jumonji et al. (2023) to make the comparison fairer.

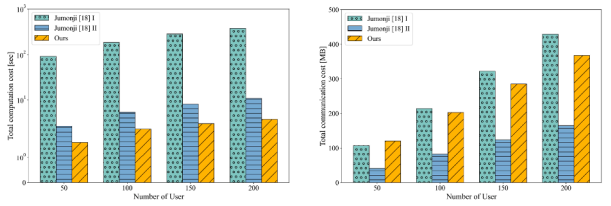
Note that the work of Jumonji et al. (2023) computes the similarity by first performing a multiplication over the packed ciphertexts and then evaluating homomorphic rotations. However, by introducing additional re-encryption, we use coefficient embedding to avoid the numerous homomorphic rotations. Since the homomorphic re-encryption is performed once for each user in our protocol while the rotations are performed at least $O(\log(M))$ times for each user in the work of Jumonji et al. (2023). Thus, theoretically, our similarity computation process is more efficient than that of Jumonji et al. (2023). The experiments support the above theoretical analysis. As illustrated in Table 5, our overall similarity computation process is about 8× more efficient than the SEAL implementation of Jumonji et al. (2023) and even much more efficient than their original HElib implementation.

Note that we have the additional secure comparison process to achieve better security. Therefore, we have additional performance loss in terms of efficiency. However, as Fig. 5 shows, our protocol outperforms the work of Jumonji et al. (2023) with respect to different numbers of users. This is because the secure comparison process takes less time than the similarity computation process, as in Fig. 3. However, our protocol consumes more communication overheads. This is because

Table 5
The costs of similarity computation process.

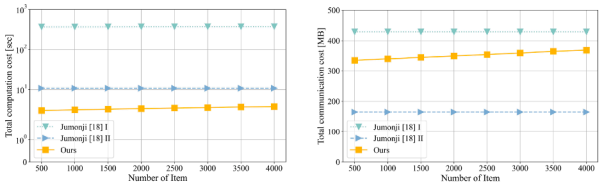
Protocols \ Process	(N-1)Mul (s)	Rotations(s)	Re-encryption	All (s)
Jumonji et al. (2023) I	11.01	322.09	–	333.10
Jumonji et al. (2023) II	0.81	8.31	–	9.12
Ours	0.81	–	1.07	**1.88**

The time cost is computed for the case of 200 users and 4000 items.



(a) Comparisons of time. (b) Comparisons of communication.

Fig. 5. Performance comparisons with respect to different numbers of users.



(a) Comparisons of time. (b) Comparisons of space.

Fig. 6. Performance comparisons with the number of items.

the re-encryption process and finding top-k predictions need additional communications.

Observe that the performance of our protocol can be influenced by the number of items, while the previous work is not if the number of items is smaller than the packing size. This is because our protocol needs additional computation and communications, which is linear to the number of items. Fig. 6 compares the time and space costs with respect to the number of items in the recommendation system when the number of users is 200. Note that our protocol still outperforms the previous state-of-the-art protocol of Jumonji et al. (2023) in time cost. However, according to the theoretical analysis, the space cost of our protocol is somewhat inefficient. This is because of the additional transactions generated during the re-encryption and comparisons.

7. Conclusion

This paper proposed a privacy-preserving collaborative filtering protocol. Although the proposed protocol can protect the predictions from the target user, the communication overhead increases the number of items. Thus, an interesting research direction is designing a privacy-preserving collaborative filtering protocol that achieves prediction privacy against all participants and has significantly smaller communication overheads. Despite the proposed protocol achieving better computation efficiency, the similarity computation process brings more communication costs. Therefore, improving the communication cost of the similarity computation process with at least as much efficiency as the proposed protocol left further work. Note that the proposed protocol maintains the target user's full rating information, while the previous works do not. Thus, maintaining both the user's rating information is also worth studying. From the implementation point of view, the high computational and communication costs associated with techniques like homomorphic encryption may hinder their application in large-scale recommendation systems. Therefore, balancing user privacy protection with the efficiency and accuracy of recommendation systems remains a critical area for further exploration.

CRediT authorship contribution statement

Parhat Abla: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing; **Jun Guang Yao:** Resources, Software, Visualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Supplementary material

Supplementary material associated with this article can be found in the online version at 10.1016/j.eswa.2026.133423.

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